

MAT1002 Lecture 26,

Outline

- Stokes' theorem
- Divergence and the Divergence Theorem

Stokes' Theorem

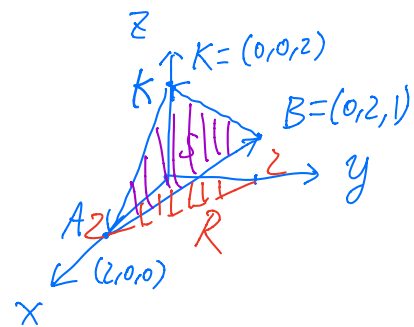
Let us start with an example.

Example

Compute the line integral of the field $\mathbf{F} := \langle x + 2y, x + 3z, 2x + y \rangle$ along C , where C consists of the edges of the triangle with vertices $A = (2, 0, 0)$, $B = (0, 2, 1)$ and $K = (0, 0, 2)$, traversing in the order of $ABKA$.

• $C = BK \cup KA \cup AB$

• $BK: \vec{r}(t) = \langle 0, 2, 1 \rangle + t\vec{BK}$
 $= \langle 0, 2-2t, 1+t \rangle, 0 \leq t \leq 1$



$$\int_{BK} \vec{F} \cdot d\vec{r} = \int_{BK} Mdx + Ndy + Pdz$$

$$= \int_0^1 0dt + 3(1+t)(-2)dt + (2-2t)dt$$

$$= \int_0^1 (-4-8t)dt = (-4t-4t^2)\Big|_0^1 = -8$$

• Need to do this for KA and AB as well, tedious...

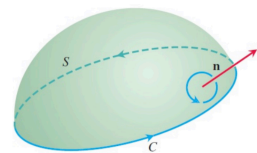
Stoke's theorem gives another way to find it.

Theorem (Stokes' Theorem) *orientable, with a side given by $\vec{n}$*

Let S be a piecewise smooth oriented surface having a piecewise smooth boundary curve C , and let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region containing S . If C is positively oriented (counterclockwise) with respect to the surface's unit normal $\mathbf{n}$, then

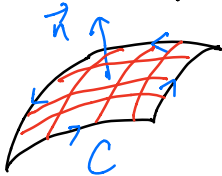
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl} \mathbf{F} \cdot \mathbf{n} d\sigma.$$

if $\vec{n}$ is "up", then C is counterclockwise when viewed from above



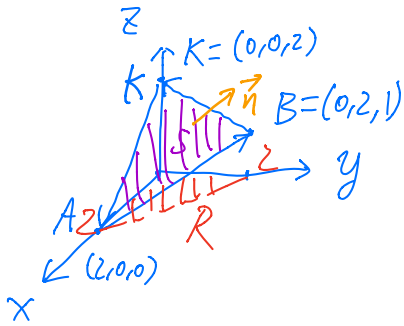
Remarks

- $\text{curl } \vec{F}$ measures rotation behaviours caused by $\vec{F}$. At a point P_0 :
 - ↳ Direction of $\text{curl } \vec{F}$ indicates positive axis of rotation of a "very very small" sphere centered at P_0 . 
 - ↳ Length measures "how fast" the small sphere rotates.
- $(\text{curl } \vec{F} \cdot \vec{n})(P_0)$ gives the circulation density on a tiny plane containing P_0 with unit normal $\vec{n}$. *Counterclockwise w.r.t. upward normal $\vec{n}$*
- Green's thm is a special case of Stokes' theorem, where C and S are entirely on the xy -plane and $\vec{n} = \vec{k} = \langle 0, 0, 1 \rangle$.
- Intuition of Stokes' theorem:



$$\begin{aligned} & \text{Total circulation around } C \\ &= \sum \text{circulation around small closed curve} \quad \text{as in Green's thm} \\ & \quad \text{(due to cancellation)} \\ &\approx \sum \text{circulation around small closed plane curve} \\ &= \sum (\text{curl } \vec{F} \cdot \vec{n}) \Delta \sigma \end{aligned}$$

Exact form: $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, d\sigma$



Let us re-do the question using Stoke's thm.

$\vec{F} = \langle x+2y, x+3z, 2x+y \rangle$, find

$$\oint_C \vec{F} \cdot \vec{T} ds.$$

Sol. • C is counterclockwise when $\vec{n}$ is "upward" (z -component > 0).

• Upward normal to plane is

$$\vec{AB} \times \vec{AK} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & 2 & 1 \\ -2 & 0 & 2 \end{vmatrix} = \langle 4, 2, 4 \rangle$$

$$\cdot \vec{n} = \langle 4, 2, 4 \rangle \frac{1}{\sqrt{4^2+2^2+4^2}} = \langle \frac{2}{3}, \frac{1}{3}, \frac{2}{3} \rangle$$

$$\cdot \text{Curl}(\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y & x+3z & 2x+y \end{vmatrix} = \langle 1-3, 0-2, 1-2 \rangle = \langle -2, -2, -1 \rangle$$

$$\cdot S: 4(x-z) + 2y + 4z = 0, \quad \text{above } R$$

$$\Leftrightarrow z = \underbrace{2-x-\frac{1}{2}y}_f, \quad \text{above } R$$

$$\cdot |\vec{r}_x \times \vec{r}_y| = \sqrt{f_x^2 + f_y^2 + 1} = \sqrt{1 + \frac{1}{4} + 1} = \frac{3}{2}$$

$$\cdot d\sigma = |\vec{r}_x \times \vec{r}_y| dx dy = \frac{3}{2} dx dy$$

$$\cdot \oint_C \vec{F} \cdot \vec{T} ds = \iint_S \text{Curl}(\vec{F}) \cdot \vec{n} d\sigma = \iint_R \langle -2, -2, -1 \rangle \cdot \langle \frac{2}{3}, \frac{1}{3}, \frac{2}{3} \rangle \frac{3}{2} dx dy$$

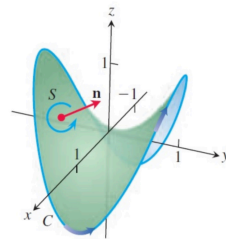
$$= \iint_R (-2+(-1)+(-1)) dA = -4 \iint_R dA = -4 \cdot \text{Area}(R)$$

$$= (-4) \frac{1}{2} \cdot 2 \cdot 2 = -8.$$

e.g. Let S be the surface $z = y^2 - x^2$ inside the cylinder $x^2 + y^2 \leq 1$.

Find upward flux of $\text{curl}(\vec{F})$ across S , where $\vec{F} = \langle y, -x, x^2 \rangle$.

Sol: • Flux = $\iint_S (\text{curl}(\vec{F}) \cdot \vec{n}) d\sigma = \oint_C \vec{F} \cdot d\vec{r}$, ^{Stokes'}



where $C = \{(x, y, y^2 - x^2) : x^2 + y^2 = 1\}$

$= \{(r \cos \theta, r \sin \theta, r^2(\sin^2 \theta - \cos^2 \theta)) : r=1, 0 \leq \theta \leq 2\pi\}$

$= \{(\cos \theta, \sin \theta, \sin^2 \theta - \cos^2 \theta) : 0 \leq \theta \leq 2\pi\}$

$\vec{r}(\theta)$, counterclockwise from above

• $\oint_C \vec{F} \cdot d\vec{r} = \oint_C M dx + N dy + P dz$

$= \int_0^{2\pi} \sin \theta (-\sin \theta d\theta) + (-\cos \theta) \cos \theta d\theta + \cos^2 \theta (2 \sin \theta \cos \theta + 2 \cos \theta \sin \theta) d\theta$

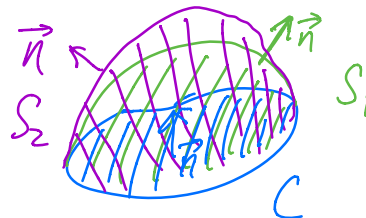
$= \int_0^{2\pi} (-1 + 4 \cos^2 \theta \sin \theta) d\theta = [-\theta - \cos^4 \theta]_{\theta=0}^{2\pi} = -2\pi$

A Consequence of Stokes' Theorem

If S_1 and S_2 are two oriented surfaces having the same boundary curve, ~~which is counterclockwise~~, ^{← having $\vec{n}$ pointing the same side} then Stokes' theorem implies that

that is
a simple
closed
curve

$$\iint_{S_1} \text{curl} \mathbf{F} \cdot \mathbf{n} d\sigma = \iint_{S_2} \text{curl} \mathbf{F} \cdot \mathbf{n} d\sigma.$$



e.g. Find circulation of $\vec{F} = \langle x^2z, y^2+zx, z^2-y \rangle$ along C ,
 where C is intersection of $S_1: x^2+y^2+z^2=1$ and $S_2: z=\sqrt{x^2+y^2}$,
 counterclockwise when viewed from above. (e.g. 16.7.7)

Sol: • ①, ② $\Rightarrow x^2+y^2+(x^2+y^2)=1 \Rightarrow x^2+y^2=\frac{1}{2}$
 $\Rightarrow C$ is circle of radius $1/\sqrt{2}$, in plane $z=1/\sqrt{2}$.

• Can take surface S lying on the plane; i.e.,

$$S = \{(x, y, \frac{1}{\sqrt{2}}) : x^2+y^2 \leq \frac{1}{2}\}.$$

• Take upward normal $\vec{n} = \vec{k}$.

• By Stokes' thm,

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \iint_S \text{curl} \vec{F} \cdot \vec{k} \, d\sigma = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) d\sigma \\ &= \iint_S (2-0) \, d\sigma = 2 \text{Area}(S) = 2\pi \frac{1}{2} = \pi. \end{aligned}$$

3D Component Test

If $\vec{F} = \langle M, N, P \rangle$, M, N, P have cts partials on $D \subseteq \mathbb{R}^3$ (D open, simply connected), then

$$\vec{F} \text{ is conservative on } D \Leftrightarrow \begin{matrix} P_y = N_z, & M_z = P_x, & N_x = M_y & \text{on } D \\ \text{①} & & \text{②} \end{matrix}$$

Proof: Note that ② is the same as $\text{curl } \vec{F} \equiv \vec{0}$ on D .

① $\Rightarrow$ ②: Assume ①. Then $\vec{F} = \langle f_x, f_y, f_z \rangle$ for some f on D .

Now $P_y = f_{zy} \underset{\text{Mixed Derivative Thm}}{=} f_{yz} = N_z$. Similarly, $M_z = P_x$, $N_x = M_y$.

② $\Rightarrow$ ①: Let C be any closed curve in D . By Stokes' thm,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, d\sigma.$$

But $\text{curl } \vec{F} \equiv \vec{0}$ on D by ②, so $\text{RHS} = 0$; i.e. $\oint_C \vec{F} \cdot d\vec{r} = 0$.

D satisfies the loop property, so it is conservative on D . □

Intuitively, $\text{curl } \vec{F} \equiv \vec{0}$ on D means the field $\vec{F}$ causes no rotation at every point; such a field is called **irrotational**. Hence, the component test states that **conservative fields and irrotational fields are the same thing (on an open simply connected domain)**.

Divergence

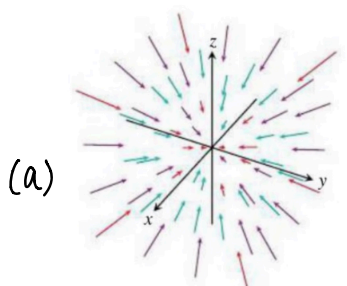
Def: Given a vector field $\vec{F}$, the **divergence of $\vec{F}$** is
 $\text{div}(\vec{F}) := \nabla \cdot \vec{F}$.

- If $\vec{F} = \langle M, N, P \rangle$, then $\text{div } \vec{F} = \frac{\partial}{\partial x}M + \frac{\partial}{\partial y}N + \frac{\partial}{\partial z}P$.
- If $\vec{F} = \langle M, N \rangle$, then $\text{div } \vec{F} = \frac{\partial}{\partial x}M + \frac{\partial}{\partial y}N$.

Intuitively, $\text{div}(\vec{F})$ at P describes the outward flux density at P , as discussed in the section of Green's thm (except now it is outward flux divided by volume if $\vec{F}$ is a 3D field).

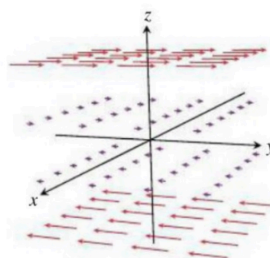
$$\text{div}(\vec{F})(P) \begin{cases} > 0 \Rightarrow \text{expanding (diverging) at } P \\ < 0 \Rightarrow \text{compressing (shrinking) at } P \\ = 0 \Rightarrow \text{neither.} \end{cases}$$

e.g.



$$\mathbf{F}(x, y, z) = -x\mathbf{i} - y\mathbf{j} - z\mathbf{k}$$

(b)



$$\mathbf{F}(x, y, z) = z\mathbf{j}$$

$$(a) \text{div}(\vec{F}) = M_x + N_y + P_z = -1 - 1 - 1 = -3. \quad (\text{Compressing})$$

$$(b) \text{div}(\vec{F}) = M_x + N_y + P_z = 0 + 0 + 0 = 0.$$

The following is a property of divergence and curl.

Theorem

If $\mathbf{F}$ is a vector field in $\mathbb{R}^3$ whose components have continuous second partial derivatives, then

$$\text{div}(\text{curl } \mathbf{F}) \equiv 0.$$

Example

Show that $\mathbf{F}(x, y, z) := \langle xy, xyz, -y^2 \rangle$ is not the curl field of any field.

Divergence Theorem

Theorem (Divergence Theorem)

Let $\mathbf{F} = \langle M, N, P \rangle$ be a vector fields whose components have continuous partial derivatives. If E is a ^{closed} bounded solid having S as its boundary, where S is a closed, piecewise smooth surface, oriented **outward** (i.e. its unit normals point out from E), then

$$\underbrace{\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma}_{\text{outward flux}} = \iiint_E \operatorname{div} \mathbf{F} \, dV.$$

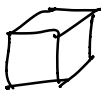
Remark

Here, roughly speaking, a closed surface is a surface that divides the space into two regions, "inside" (which is bounded) and "outside" (which is unbounded).

e.g.

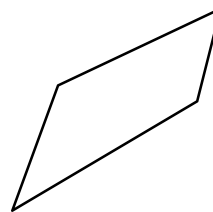


$$x^2 + y^2 + z^2 = a^2$$



cubic shell

closed



not closed